

olver C.

Sec (8.1-8.5, 8.7, 10.5)

Fundamentals of Applied Probability and Random Processes

Module 3.

- * Random process & classification
- * mean & auto correlation
- * wide sense stationary process (WSS)
- * auto correlation & power spectral density of WSS processes and their properties.
- * Poisson process - distribution of inter-arrival times.
- * Combination of independent Poisson processes (merging) & sub-division (splitting) of Poisson processes (results without proof)

Introduction

A random variable X that maps an event $\omega \in S$, where S is the sample space, to some numbers $X(\omega)$. Random processes enlarges the r.v concept to include time.

This implies that instead of the number $X(\omega)$ we deal with $X(t, \omega)$, where $t \in T$. T is called the parameter set of the process and is usually a set of times.

Classification of Random processes

A random process can be classified according to the values of $X(t, \omega)$ and the nature of the time parameter T . If T is a countable set and hence it is discrete,

the process is called a discrete time random process or a random sequence which is denoted by $\{X[n], n=1, 2, \dots\}$. If T is an interval of real numbers and hence is continuous, the process is called continuous time random process.

The values that $x(t)$ assumes are called the states of the random process. The set of all possible values of $x(t)$ forms the state space, E of the random process. If E is discrete, the process is called a

discrete state random process and if E is continuous, the process is called a continuous state random process.

Characterising a random Process.

A r.v. $X(s)$ is represented by X by suppressing the sample space parameter s .

||^{ely} a random process $X(t, s)$ is represented by $X(t)$. A ~~random~~ random

process is completely described or characterised by the joint p.d.f

$$F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) \\ = P(X(t_1) \leq x_1, X(t_2) \leq x_2, \dots, X(t_n) \leq x_n)$$

where $0 \leq t_1 < t_2 < \dots < t_n$

If $X(t)$ is continuous, then p.m.f. is

$$f_X(x_1, x_2, \dots, x_n; t_1, \dots, t_n) = \frac{\partial^n F_X(x_1, \dots, x_n; t_1, \dots, t_n)}{\partial x_1 \partial x_2 \dots \partial x_n}$$

If $X(t)$ is ~~cont~~ discrete, then p.m.f. is

$$P_X(x_1, \dots, x_n; t_1, \dots, t_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

Mean and auto correlation function of a random process.

The Mean of $x(t)$ is a function of time and is denoted by $\mu_x(t) = E(x(t))$

$$E(x(t)) = \sum_i x_i f(x_i, t) \quad ; \quad \text{For discrete state process}$$

$$E(x(t)) = \int_{-\infty}^{\infty} x f(x, t) dx \quad ; \quad \text{For continuous state process}$$

$$E(x^2(t)) = \sum_i x_i^2 f(x_i, t) \quad ; \quad \text{For discrete state process}$$

$$E(x^2(t)) = \int_{-\infty}^{\infty} x^2 f(x, t) dx \quad ; \quad \text{For continuous state process.}$$

Auto correlation of the process $x(t)$ is given by

$$R_{xx}(t_1, t_2) = E(x(t_1) \cdot x(t_2))$$

Auto correlation with period T is given by $\frac{1}{2T} \int_{-T}^T x(t) dx(t)$
Auto Covariance of the process $x(t)$ is denoted

by $C_{xx}(t_1, t_2)$. or $c(t_1, t_2)$ is defined as.

$$C_{xx}(t_1, t_2) = E \left\{ [x(t_1) - M_x(t_1)] \cdot [x(t_2) - M_x(t_2)] \right\}$$

$$\text{i.e., } C_{xx}(t_1, t_2) = E \left\{ x(t_1) \cdot x(t_2) - M_x(t_1) M_x(t_2) \right\}$$

$$= E(x(t_1) x(t_2)) - E(x(t_1)) \cdot E(x(t_2))$$

$$C_{xx}(t_1, t_2) = R_{xx}(t_1, t_2) - E(x(t_1)) \cdot E(x(t_2))$$

Correlation coefficient of the process $x(t)$ is denoted by $\rho_{xx}(t_1, t_2)$ is defined by

$$\rho_{xx}(t_1, t_2) = \frac{C_{xx}(t_1, t_2)}{\sigma_x(t_1) \cdot \sigma_x(t_2)} = \frac{C_{xx}(t_1, t_2)}{\sqrt{V_x(t_1)} \sqrt{V_x(t_2)}}$$

Variance of the process $x(t)$ is

$$\text{defined as } V(x(t)) = E(x^2(t)) - E(x(t))^2$$

$$\text{or } E[(x - \mu)^2] \text{ or } E[x(t) - E(x(t))]^2$$

problems:

1. A random process is defined by $x(t) = k \cos \omega t$ $t \geq 0$ where ω is a constant and k is uniformly distributed b/w 0 and 2. Determine
- 1) $E(x(t))$
 - 2) The autocorrelation function of $x(t)$
 - 3) The auto. covariance of $x(t)$.

Soln) Since k is uniformly distributed b/w 0 and 2 we have

$$E(k) = \frac{0+2}{2} = 1$$

$$V(k) = \frac{(2-0)^2}{12} = \frac{1}{3}$$

$$E(k^2) = V(k) + [E(k)]^2$$

$$= \frac{1}{3} + 1 = \frac{4}{3}$$

$$f(x) = \frac{1}{\beta - \alpha}$$

$$V(x) = E(x^2) - (E(x))^2$$

$$E(x) = \frac{\alpha + \beta}{2}$$

$$V(x) = \frac{(\beta - \alpha)^2}{12}$$

$$E(x(t)) = E(k \cos \omega t) = E(k) \cos \omega t$$

$$= \cos \omega t$$

2. The autocorrelation function of $x(t)$ is

$$R_{xx}(t_1, t_2) = E(x(t_1)x(t_2))$$

$$((0, \pi) \Rightarrow) E(k \cos \omega t, k \cos \omega t_2)$$

$$((0, \pi) \Rightarrow) E(k^2) \cos \omega t, \cos \omega t_2$$

$$((0, \pi) \Rightarrow) \frac{4}{3} \cos \omega t, \cos \omega t_2.$$

3. The auto covariance of $x(t)$ is

$$C_{xx}(t_1, t_2) = R_{xx}(t_1, t_2) - E(x(t_1))E(x(t_2))$$

$$= \frac{4}{3} \cos \omega t_1 \cos \omega t_2 - \cos \omega t_1 \cos \omega t_2$$

$$= \frac{4}{3} \cos \omega t_1 \cos \omega t_2 - \cos \omega t_1 \cos \omega t_2$$

$$= \left(\frac{4}{3} - 1 \right) \cos \omega t_1 \cos \omega t_2 = \frac{1}{3} \cos \omega t_1 \cos \omega t_2$$

2. Consider a random process $x(t)$ defined by

$x(t) = A \cos(\omega t + \theta)$ where A and θ are independent and uniform random variable over $(-k, k)$ and $(-\pi, \pi)$ respectively. Find

- 1) Mean of $x(t)$
- 2) Auto correlation function of $x(t)$
- 3) Auto covariance function of $x(t)$
4. Variance of $x(t)$.

8th) 1) Mean of $x(t) = E(x(t))$

$$= E(A \cos(\omega t + \theta))$$

$$= E(A) E(\cos(\omega t + \theta))$$

$$= 0 \times E(\cos(\omega t + \theta))$$

$$= \underline{\underline{0}}$$

$$A \sim U(-k, k)$$

$$f(A) = \frac{1}{2k} \quad -k \leq A \leq k$$

$$f(\theta) = \frac{1}{2\pi}, \quad -\pi \leq \theta \leq \pi$$

Uniform distribution

$$\text{Mean} = \frac{\alpha + \beta}{2}$$

$$\alpha = -k \quad \beta = k$$

$$\text{Mean} = 0$$

$$E(A) = 0$$

$$\text{Similarly } E(\theta) = 0$$

2) Auto correlation function of $x(t)$

$$R_{xx}(t_1, t_2) = E(x(t_1) x(t_2))$$

$$= E(A \cos(\omega t_1 + \theta) A \cos(\omega t_2 + \theta))$$

$$= E(A^2 \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta))$$

$$= E(A^2) E(\cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)) \quad \text{--- (1)}$$

$$E(A^2) = \int_{-k}^k A^2 f(A) dA$$

$$= \int_{-k}^k A^2 \frac{1}{2k} dA = \frac{1}{2k} \left[\frac{A^3}{3} \right]_{-k}^k$$

$$= \frac{1}{6k} [k^3 - (-k)^3] = \frac{2k^3}{6k} = \frac{k^2}{3} \quad \text{--- (2)}$$

$$E[\cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)]$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\cos(A-B) + \cos(A+B) = 2 \cos A \cos B$$

$$\cos A \cos B = \frac{\cos(A-B) + \cos(A+B)}{2}$$

$$\text{Here } A = \omega t_1 + \theta \quad B = \omega t_2 + \theta$$

$$\begin{aligned} \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) &= \frac{\cos(\omega t_1 - \omega t_2) + \cos(\omega t_1 + \omega t_2 + 2\theta)}{2} \\ &= \frac{\cos \omega(t_1 - t_2) + \cos(\omega(t_1 + t_2) + 2\theta)}{2} \end{aligned}$$

$$E(\cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)) = \int_{-\pi}^{\pi} \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) f(\theta) d\theta$$

$$= \int_{-\pi}^{\pi} \left[\frac{1}{2} \{ \cos \omega(t_1 - t_2) + \cos(\omega(t_1 + t_2) + 2\theta) \} \right] \frac{1}{2\pi} d\theta$$

$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} [\cos \omega(t_1 - t_2) + \cos(\omega(t_1 + t_2) + 2\theta)] d\theta$$

$$= \frac{1}{4\pi} \left[\cos \omega(t_1 - t_2) \theta + \frac{\sin(\omega(t_1 + t_2) + 2\theta)}{2} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{4\pi} \left[\cos \omega(t_1 - t_2) 2\pi + \frac{\sin(\omega(t_1 + t_2) + 2\pi)}{2} \right]$$

$$= \frac{1}{4\pi} \left[\cos \omega(t_1 - t_2) 2\pi + \frac{\sin(\omega(t_1 + t_2) - 2\pi)}{2} \right]$$

$$= \frac{1}{4\pi} \left[\frac{\sin \omega(t_1+t_2)}{2} + \pi \cos \omega(t_1-t_2) \right.$$

$$\left. - \frac{\sin(-2\pi - \omega(t_1+t_2))}{2} + \pi \cos \omega(t_1-t_2) \right]$$

$$= \frac{1}{4\pi} \left[\frac{\sin \omega(t_1+t_2)}{2} + \sin(2\pi - \omega(t_1+t_2)) + \right.$$

$$\left. 2\pi \cos \omega(t_1-t_2) \right]$$

$\sin(2\pi - \theta) = -\sin \theta$

$$= \frac{1}{4\pi} \left[\frac{\sin \omega(t_1+t_2)}{2} - \frac{\sin \omega(t_1+t_2)}{2} + 2\pi \cos \omega(t_1-t_2) \right]$$

$$= \frac{1}{4\pi} [2\pi \cos \omega(t_1-t_2)]$$

$$= \cos \omega(t_1-t_2) \quad \text{--- (3)}$$

Substitute eq (1), (2) & (3) in eq (1)

Auto correlation $R_{xx}(t_1, t_2) = \frac{k^2}{\sigma_x^2} \cdot \frac{\cos \omega(t_1-t_2)}{2}$

$$= \frac{k^2}{\sigma_x^2} \cos \omega(t_1-t_2)$$

3. Auto Covariance of $x(t) = e_{xx}(t_1, t_2)$

$$R_{xx}(t_1, t_2) = E[x(t_1)]E[x(t_2)]$$

$$= \frac{k^2}{6} \cos[\omega(t_1 - t_2)] - E[x(t_1)] E[x(t_2)]$$

$$E[x(t_1)] = E[A \cos(\omega t_1 + \theta)] \\ = E(A) E(\cos \omega t_1 + \theta) \\ = 0$$

$$E[x(t_2)] = E[A \cos(\omega t_2 + \theta)] \\ = E(A) E(\cos \omega t_2 + \theta) \\ = 0$$

Auto covariance of $x(t) = c_{xx}(t_1, t_2) =$

$$\frac{k^2}{6} \cos[\omega(t_1 - t_2)] - 0$$

$$= \frac{k^2}{6} \cos[\omega(t_1 - t_2)]$$

4 Variance of $x(t) = v(x(t)) = E[x(t) - E(x(t))]^2$

$$= E[x(t) - 0]^2$$

$$= E[x(t)^2]$$

$$= E[x(t) x(t)]$$

$$= R_{xx}(t, t)$$

$$= \frac{k^2}{6} \cos \omega(t - t)$$

$$= \frac{k^2}{6} \cos 0 = \frac{k^2}{6}$$

H-w. Consider a random process $x(t) = B \cos(50t + \phi)$ where B and ϕ are independent random variables. B is a uniform variable with mean 0 and variance 1. ϕ is uniformly distributed in the interval $[-\pi, \pi]$. Find mean and auto correlation of the process.

Soln)

$$E(B) = 0 \quad V(B) = 1$$

$$V(B) = E(B^2) - [E(B)]^2$$

$$E(B^2) = V(B) + [E(B)]^2 = 1 + 0 = 1$$

$$\begin{aligned} \text{Mean of } x(t) &= E[x(t)] = E[B \cos(50t + \phi)] \\ &= E(B) E(\cos(50t + \phi)) \\ &= 0 \end{aligned}$$

$$\text{Auto correlation of } x(t) = R_{xx}(t_1, t_2) =$$

$$E[x(t_1) x(t_2)]$$

$$= E[B \cos(50t_1 + \phi) B \cos(50t_2 + \phi)]$$

$$= E[B^2 \cos(50t_1 + \phi) \cos(50t_2 + \phi)]$$

$$= E(B^2) E(\cos(50t_1 + \phi) \cos(50t_2 + \phi))$$

$$= 1 \times E(\cos(50t_1 + \phi) \cos(50t_2 + \phi))$$

$$= E(\cos(50t_1 + \phi) \cos(50t_2 + \phi))$$

$$= \int_{-\pi}^{\pi} \cos(50t_1 + \phi) \cos(50t_2 + \phi) f(\phi) d\phi = \cos(50t_1 - 50t_2)$$

Wide sense stationary process (WSS)

Random processes in which the mean and auto correlation function do not depend on absolute time are called wide sense stationary (WSS) processes.

A random process $x(t)$ is said to be WSS if there are 2 conditions are satisfied

① Mean = $E(x(t)) = \text{Constant}$

② Auto correlation = $R_{xx}(t_1, t_2)$ must depend

only on time difference $\tau = t_1 - t_2$

i.e., $R_{xx}(t_1, t_2) = R_{xx}(t_1 - t_2)$

Problems

1. Consider the random process $x(t) = A \cos(\omega t + \theta)$ given A and ω are constants and θ is uniformly distributed in $(0, 2\pi)$. P.T $x(t)$ is WSS

Given $x(t) = A \cos(\omega t + \theta)$

and θ is uniformly distributed on $(0, 2\pi)$

$\therefore f(\theta) = \frac{1}{2\pi} \quad 0 < \theta < 2\pi$

Mean $= E(x(t)) = E(A \cos(\omega t + \theta))$

$= \int_0^{2\pi} A \cos(\omega t + \theta) f(\theta) d\theta$

$= \int_0^{2\pi} A \cos(\omega t + \theta) \frac{1}{2\pi} d\theta$

$= \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega t + \theta) d\theta$

$= \frac{A}{2\pi} \left[\sin(\omega t + \theta) \right]_0^{2\pi}$

$= \frac{A}{2\pi} \left[\sin(\omega t + 2\pi) - \sin \omega t \right]$

$= \frac{A}{2\pi} \left[\sin \omega t - \sin \omega t \right] \quad \left[\because \sin(2\pi + \theta) = \sin \theta \right]$

$= 0$

$\therefore E(x(t))$ is constant

(2) $R_{xx}(t_1, t_2) = E[x(t_1)x(t_2)]$

$= E[A \cos(\omega t_1 + \theta) A \cos(\omega t_2 + \theta)]$

$$x(t) = E[A^2 \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)]$$

$$= \int_0^{2\pi} A^2 \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) f(\theta) d\theta$$

$$= \int_0^{2\pi} A^2 \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) \frac{1}{2\pi} d\theta$$

$$= \frac{A^2}{2\pi} \int_0^{2\pi} \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) d\theta$$

$$= \frac{A^2}{2\pi} \int_0^{2\pi} \frac{[\cos \omega(t_1 + t_2) + 2\theta] + \cos \omega(t_1 - t_2)}{2} d\theta$$

$$= \frac{A^2}{2\pi} \times 2 \int_0^{2\pi} \left[\frac{\cos \omega(t_1 + t_2) + 2\theta}{2} + \cos \omega(t_1 - t_2) \right] d\theta$$

$$= \frac{A^2}{4\pi} \left[\frac{\sin[\omega(t_1 + t_2) + 2\theta]}{2} + (\cos \omega(t_1 - t_2)) \theta \right]_0^{2\pi}$$

$$= \frac{A^2}{4\pi} \left[\frac{\sin[\omega(t_1 + t_2) + 4\pi]}{2} + 2\pi \cos \omega(t_1 - t_2) \right]$$

$$= \frac{A^2}{4\pi} \left[\frac{\sin \omega(t_1 + t_2)}{2} + 2\pi \cos \omega(t_1 - t_2) \right]$$

$$= \frac{A^2}{4\pi} \left[\cancel{\sin \omega(t_1 + t_2)} + 2\pi \cos \omega(t_1 - t_2) - \frac{\cancel{\sin \omega(t_1 + t_2)}}{2} \right]$$

$$= \frac{A^2}{4\pi} \times 2\pi \cos \omega(t_1 - t_2) = \frac{A^2}{2} \cos \omega(t_1 - t_2)$$

$\therefore$ Auto correlation depends only on time difference.

$\Rightarrow$ All conditions are satisfied.

$\therefore \underline{\underline{X(t) \text{ is WSS.}}}$

2) Examine whether the Poisson process

$X(t)$ given by $p(X(t)=r) = \frac{e^{-\lambda t} (\lambda t)^r}{r!}$,

$r=0,1,2,\dots$ is covariance stationary

Soln) Given $X(t)$ is Poisson distribution with parameter λt

Mean = $E(X(t)) = \lambda t \neq \text{Constant}$

$\therefore X(t)$ is not a W.S.S.

$\therefore$ Poisson process $X(t)$ is not-covariance stationary.

A.W.

3) Let the random process defined by $X(t) = A \sin(\omega t + \theta)$ where A and θ are

constants and θ is a r.v. uniformly distributed in $[0, 2\pi]$ is a w.s.s. process.

soln) Since θ is uniformly distributed in $[0, 2\pi]$, then $f(\theta) = \frac{1}{2\pi}$ ($0 < \theta < 2\pi$)

$$R(x(t)) = E(A \sin(\omega t + \theta))$$

$$= A E(\sin(\omega t + \theta))$$

$$= A \int_0^{2\pi} \sin(\omega t + \theta) f(\theta) d\theta$$

$$= A \int_0^{2\pi} \sin(\omega t + \theta) \frac{1}{2\pi} d\theta$$

$$\cos(\theta + \pi) = -\cos \theta$$

$$= \frac{A}{2\pi} \int_0^{2\pi} \sin(\omega t + \theta) d\theta$$

$$= \frac{-A}{2\pi} \left[\cos(\omega t + \theta) \right]_0^{2\pi}$$

$$= \frac{-A}{2\pi} \left[\cos(\omega t + 2\pi) - \cos(\omega t) \right]$$

$$= \frac{-A}{2\pi} \left[\cos \omega t - \cos \omega t \right] = 0$$

$$R_{xx}(t_1, t_2) = E(x(t_1) x(t_2))$$

$$= E(A \sin(\omega t_1 + \theta) A \sin(\omega t_2 + \theta))$$

$$= A^2 E(\sin(\omega t_1 + \theta) \sin(\omega t_2 + \theta))$$

$$= \int_0^{2\pi} A^2 \sin(\omega t_1 + \theta) \sin(\omega t_2 + \theta) f(\theta) d\theta$$

$$= \int_0^{2\pi} A^2 \sin(\alpha t_1 + \theta) \sin(\alpha t_2 + \theta) \frac{1}{2\pi} d\theta$$

$$= \frac{A^2}{2\pi} \int_0^{2\pi} \sin(\alpha t_1 + \theta) \sin(\alpha t_2 + \theta) d\theta$$

$$= \frac{A^2}{2\pi} \left[\frac{\cos(\alpha t_1 - \alpha t_2)}{2} - \frac{\cos(\alpha t_1 + \alpha t_2 + 2\theta)}{2} \right]_0^{2\pi}$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$= \frac{A^2}{4\pi} \left[\cos(\alpha t_1 - \alpha t_2) (2\pi) - \frac{\sin(\alpha(t_1+t_2)+2\theta)}{2} \right]_0^{2\pi}$$

$$= \left[\frac{A^2}{4\pi} \left[\cos(\alpha t_1 - \alpha t_2) 2\pi - \frac{\sin(\alpha(t_1+t_2)+2\theta)}{2} \right] \right]_0^{2\pi}$$

$$= \frac{A^2}{4\pi} \left[2\pi \cos(\alpha t_1 - \alpha t_2) - \frac{\sin \alpha(t_1+t_2)}{2} \right]_0^{2\pi}$$

$$= \frac{2\pi A^2}{4\pi} \cos(\alpha t_1 - \alpha t_2) = \frac{A^2}{2} \cos \alpha(t_1 - t_2)$$

$\therefore$ Auto correlation depends only on
time difference.

$\Rightarrow$ All conditions are satisfied.

$\therefore$ $x(t)$ is WSS.

Auto correlation and power spectral density of wss process and their properties.

The power spectral density of a wss process $x(t)$ is defined as a fourier transform of the auto correlation function $R_{xx}(z)$ and is denoted by $S_{xx}(\omega)$

Thus

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(z) e^{-i\omega z} dz = F(R_{xx}(z))$$

(fourier transform of $R_{xx}(z)$)

$$R_{xx}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) e^{i\omega z} d\omega$$

Properties

1. $S_{xx}(\omega) \geq 0$, is a nonnegative function.
2. $S_{xx}(-\omega) = S_{xx}(\omega)$, $S_{xx}(\omega)$ is an even function.
3. The power spectral density is a real function if $x(t)$ is real.

4. The average power of $x(t)$ is given by

$$E[x^2(t)] = R_{xx}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) d\omega$$

5. $S_{xx}(\omega)$ is not a complex function

Note. - For 2 Random processes $x(t)$ and $y(t)$ that are jointly WSS, the Fourier transform of their cross correlation function $R_{xy}(z)$ is

called the cross power spectral density $S_{xy}(\omega)$ of the two random processes.

$$\text{Thus } S_{xy}(\omega) = \int_{-\infty}^{\infty} R_{xy}(z) e^{-i\omega z} dz$$

Cross power spectral density is a complex function when both $x(t)$ and $y(t)$ are real.

1. Find the power spectral density of a WSS random process $x(t)$ whose auto correlation function is

$$R(z) = \begin{cases} 1 - \frac{|z|}{T}, & |z| \leq T \\ 0 & |z| > T \end{cases}$$

Ans) Power spectral density of $x(t)$ is

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(z) e^{-i\omega z} dz.$$

$$= \int_{-T}^T \left(1 - \frac{|z|}{T}\right) e^{-i\omega z} dz.$$

$$= \int_{-T}^T \left(1 - \frac{|z|}{T}\right) (\cos \omega z - i \sin \omega z) dz$$

$$= \int_{-T}^T \left[\left(1 - \frac{|z|}{T}\right) \cos \omega z - i \left(1 - \frac{|z|}{T}\right) \sin \omega z \right] dz.$$

$$= \int_{-T}^T \left(1 - \frac{|z|}{T}\right) \cos \omega z dz - i \int_{-T}^T \left(1 - \frac{|z|}{T}\right) \sin \omega z dz.$$

$$= \int_{-T}^T \left(1 - \frac{|z|}{T}\right) \cos \omega z dz - i \int_{-T}^T \underbrace{\text{even} \times \text{odd fn.}}_{\text{odd fn.}}$$

$$= \int_{-T}^T \left(1 - \frac{|z|}{T}\right) \cos \omega z dz - 0.$$

$$= 2 \int_0^T \left(1 - \frac{z}{T}\right) \cos \omega z dz.$$

$$\begin{aligned}
& 2 \int_0^T \left(\left(1 - \frac{z}{T}\right) \left(\frac{\sin \omega z}{\omega}\right) - \left(-\frac{1}{T}\right) \left(-\frac{\cos \omega z}{\omega^2}\right) \right) dz \\
&= 2 \int_0^T \left(\left(1 - \frac{z}{T}\right) \frac{\sin \omega z}{\omega} - \frac{1}{T} \frac{\cos \omega z}{\omega^2} \right) dz \\
&= 2 \int_0^T \left(0 - \frac{\cos \omega T}{\omega^2} - \left(\frac{\sin \omega T}{\omega} - \frac{1}{T} \frac{\cos \omega \times 0}{\omega^2} \right) \right) dz \\
&= 2 \int_0^T \left(-\frac{\cos \omega T}{\omega^2} - \frac{\sin \omega T}{\omega} + \frac{1}{T \omega^2} \right) dz \\
&= 2 \int_0^T \left(-\frac{\cos \omega T}{\omega^2} + \frac{1}{T \omega^2} \right) dz
\end{aligned}$$

2. The auto correlation of the random telegraph signal process is given by

$$R(z) = a^2 e^{-2\alpha|z|}$$

Determine the process density spectrum of the random telegraph signal.

$$\text{psd is } S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(z) e^{-i\omega z} dz$$

$$= \int_{-\infty}^{\infty} a^2 e^{-2\alpha|z|} e^{-i\omega z} dz$$

$$= a^2 \int_{-b}^b e^{-2\alpha|z|} e^{-i\omega z} dz$$

$$= a^2 \int_{-b}^0 e^{-2\alpha(-z)} e^{-i\omega z} dz + a^2 \int_0^b e^{-2\alpha z} e^{-i\omega z} dz$$

$$= a^2 \int_{-b}^0 e^{+2\alpha z - i\omega z} dz + a^2 \int_0^b e^{-2\alpha z - i\omega z} dz$$

$$= a^2 \int_{-b}^0 e^{z(2\alpha - i\omega)} dz + a^2 \int_0^b e^{z(-2\alpha - i\omega)} dz$$

$$= a^2 \left[\frac{e^{z(2\alpha - i\omega)}}{2\alpha - i\omega} \right]_{-b}^0 + a^2 \left[\frac{e^{z(-2\alpha - i\omega)}}{-(-2\alpha - i\omega)} \right]_0^b$$

$$= a^2 \left[\frac{1}{2\alpha - i\omega} - 0 \right] + a^2 \left[0 - \frac{1}{-(2\alpha + i\omega)} \right]$$

$$= \frac{a^2}{2\alpha - i\omega} + \frac{a^2}{2\alpha + i\omega}$$

$$= a^2 \left[\frac{1}{2\alpha - i\omega} + \frac{1}{2\alpha + i\omega} \right]$$

$$= a^2 \left[\frac{(2\alpha + i\omega) + 2\alpha - i\omega}{(2\alpha - i\omega)(2\alpha + i\omega)} \right]$$

$$= a^2 \left[\frac{4\alpha}{4\alpha^2 + \omega^2} \right] = \frac{a^2 4\alpha}{4\alpha^2 + \omega^2}$$

H.W

3. Q.

Find PSD of the ^{wes} process $x(t)$ with auto correlation fn $R_x(z) = \begin{cases} 1-|z| & -1 \leq z \leq 1 \\ 0 & \text{otherwise} \end{cases}$

4. Q. Determine the auto correlation function of the random process with the power spectral density

$$S_{xx}(\omega) = \begin{cases} S_0 & |\omega| \leq \omega_0 \\ 0 & \text{otherwise} \end{cases}$$

soln) $S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(z) \cdot e^{-i\omega z} dz$

$$R_{xx}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) e^{i\omega z} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} S_0 e^{i\omega z} d\omega$$

$$= \frac{S_0}{2\pi} \left[\frac{e^{i\omega z}}{iz} \right]_{-\omega_0}^{\omega_0}$$

$$= \frac{S_0}{2\pi iz} \left[e^{i\omega_0 z} - e^{-i\omega_0 z} \right]$$

$$= \frac{S_0}{\pi z} \left[\frac{e^{i\omega_0 z} - e^{-i\omega_0 z}}{2i} \right]$$

$$= \frac{S_0}{\pi z} \sin \omega_0 z$$

Poisson Process.

Let $x(t)$ be the counting process which represents the number of occurrences of a certain event in the interval $(0, t)$. Then $x(t)$ is called a Poisson process with rate or intensity λ if it satisfies the following conditions.

- 1) $x(0) = 0$
- 2) $x(t)$ has independent and stationary increments

3) For any t , $x(t) \sim \text{Poisson}(\lambda t)$

$$p(x(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, n = 0, 1, 2, \dots$$

Note: I

If the question is about no. of arrivals then only apply poisson probability, where

$$p(x|t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, n=0,1,2,\dots$$

Note 2.

If the question is about time b/w arrivals or inter arrival time then apply exponential probability where $f(t) = \lambda e^{-\lambda t}$.

Problems

1. P.T the poisson process is not a.s.s
(or stationary process)

Ans) For a random process to be stationary its mean must be constant. But for a

poisson process mean = λt which is not constant. Hence poisson process is not a stationary process.

2. Q Suppose that the customers arrive at a bank counter in accordance with a poisson process with mean rate λ , 3 per hour. Find the probability that in the next 2 hrs

- 1) exactly 4 customers arrive.
- 2) fewer than 4 customers arrive.
- 3) atleast 4 customers arrive

Soln) given mean rate $\lambda = 3$ per hr.

Here the question is about the problem of arrival.

$$p(x(t)=n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, \quad n=0,1,2,\dots$$

$$\lambda = 3, \quad t = 2.$$

$$p(x(2)=n) = \frac{e^{-3 \cdot 2} (3 \cdot 2)^n}{n!} = \frac{e^{-6} 6^n}{n!}, \quad n=0,1,2,\dots$$

$$1) \quad p[\text{exactly 4 customers arrive}] = p(x(2)=4)$$

$$= \frac{e^{-6} (6)^4}{4!} = \underline{\underline{0.1339}}$$

2) $p(\text{fewer than 4 customers arrive})$

$$\begin{aligned} & p(x(2) < 4) \\ &= p(x(2) = 0) + p(x(2) = 1) + p(x(2) = 2) \\ & \quad + p(x(2) = 3) \\ &= \frac{e^{-6} 6^0}{0!} + \frac{e^{-6} 6^1}{1!} + \frac{e^{-6} 6^2}{2!} + \frac{e^{-6} 6^3}{3!} \\ &= \underline{\underline{0.1512}} \end{aligned}$$

3) $p(\text{at least 4 customers arrive})$

$$\begin{aligned} &= p(x(2) \geq 4) \\ &= 1 - p(x(2) = 0, 1, 2, 3) \\ &= 1 - 0.1512 \\ &= \underline{\underline{0.8488}} \end{aligned}$$

3. Q. Customers arrive at a counter in accordance with Poisson process with a mean rate of 2 per minutes. Find the probability that

the time b/w two successive arrivals
(inter-arrival time) is

- 1) less than 1 minute
- 2) about 1 minutes.
- 3) more than 2 minutes
- 4) 4 minutes or less.
- 5) b/w 1 and 2 minutes.

Soln) Question is about inter arrival time

so apply exponential probability.

$$f(t) = \lambda e^{-\lambda t} = 2e^{-2t}, \quad t > 0$$

Here $\lambda = 2$

1. $p(\text{less than 1 minute}) = p(t < 1) = \int_0^1 2e^{-2t} dt$
 $= 2 \left[\frac{e^{-2t}}{-2} \right]_0^1 = -[e^{-2} - 1] = 1 - e^{-2} = 0.8647$

2. $p(\text{about 1 minute}) = p(t \geq 1) = \int_1^{\infty} 2e^{-2t} dt$
 $= 2 \left[\frac{e^{-2t}}{-2} \right]_1^{\infty} = -[e^{-\infty} - e^{-2}] = e^{-2}$
 $= 0.1353$

3) $p(\text{time more than 2 minutes})$

$$= p(t > 2)$$

$$= \int_2^{\infty} 2 e^{-2t} dt = 2 \left[\frac{e^{-2t}}{-2} \right]_2^{\infty}$$

$$= e^{-4} = \underline{\underline{0.0183}}$$

4) $p(\text{time 4 minutes or less}) = p(t \leq 4)$

$$= \int_0^4 2 e^{-2t} dt = \underline{\underline{0.9997}}$$

5) $p(\text{time between 1 + 2 minutes})$

$$= p(1 \leq t \leq 2) = \int_1^2 2 e^{-2t} dt = 2 \left[\frac{e^{-2t}}{-2} \right]_1^2$$

$$= \underline{\underline{0.1170}}$$

4.9. Car 1 arrive at a gas station according to a Poisson process at an average rate of 12 cars per hour. The station has only one attendant. If the attendant decides to take a 2 minute coffee break when there are no

cars at the station. What is the probability that one ~~or~~ or more cars will be waiting when he comes back from the break given that any car that arrives when he is on coffee break waits for him to get back?

Ans) Here the question is arrival.

$$\therefore p(x(t)=n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, \quad n = 0, 1, 2, \dots$$

$$\lambda = 12 \text{ cars per hour} = \frac{12}{60} = \frac{1}{5}$$

$$t = 2 \text{ minute}$$

$$p(x(2) \geq 1) = p(x(2)=1) + p(x(2)=2) + \dots$$

$$= 1 - p(x(2)=0)$$

$$= 1 - \left[\frac{e^{-\frac{1}{5}} \left(\frac{1}{5} \cdot 2\right)^0}{0!} \right]$$

$$= 1 - e^{-2/5} = 0.3297$$

$$\underline{\underline{0.3297}}$$

5. If the customers arrive at a counter in accordance with Poisson distribution with rate of 2 per minute. Find the probability that the interval between 2 consecutive arrivals is
 1) more than 1 minute 2) b/w 1 minute and 2 minutes.

Here the question is about time b/w arrivals.
 $\therefore$ exponential probability.

$$f(t) = \lambda e^{-\lambda t}$$

$\lambda = 2$ per minute

$$\therefore f(t) = 2e^{-2t}$$

$$1) p(\text{more than 1 minute}) = p(t > 1)$$

$$= 1 - p(t \leq 1) = 1 - \int_0^1 2e^{-2t} dt$$

$$= 1 - \left[\frac{2e^{-2t}}{-2} \right]_0^1$$

$$= 1 - [e^{-2} - 1]$$

$$= 1 - e^{-2} = 1 - 0.135$$

$$= 0.865$$

$$2) P(1 < t < 2) = \int_1^2 2e^{-2t} dt = 2 \left[\frac{e^{-2t}}{-2} \right]_1^2 = \underline{\underline{0.117}}$$